1. Problem Understanding

Description of the Problem

Modern healthcare faces the challenge of moving from reactive treatment to proactive risk management for chronic diseases like diabetes.The primary goal is to develop a reliable classification system to accurately identify "at-risk" individuals using non-invasive patient health attributes.This project compares two computational approaches—Traditional Machine Learning and Neural Networks (Deep Learning)—to determine which architecture better handles data variability and noise in indicators like BMI, age, and blood pressure to provide a binary prediction of diabetes risk.

Importance of Predicting Diabetes Risk

Predicting diabetes risk is essential for public health and well-being because it enables:

- **Early Intervention:** Allowing lifestyle changes before the condition becomes chronic.
- **Resource Optimization:** Prioritizing urgent cases to reduce healthcare infrastructure burdens with tailored insights.
- **Cost and Complication Reduction:** Lowering long-term expenses by preventing severe complications like kidney failure.
- **Enhanced Quality of Life:** Improving patient outcomes through preventative measures.
- **Targeted Public Health:** Facilitating focused screenings and awareness strategies.

2. Data Exploration Insights

Key Observations from the Dataset

Key dataset characteristics for the **6,000 patients** include:

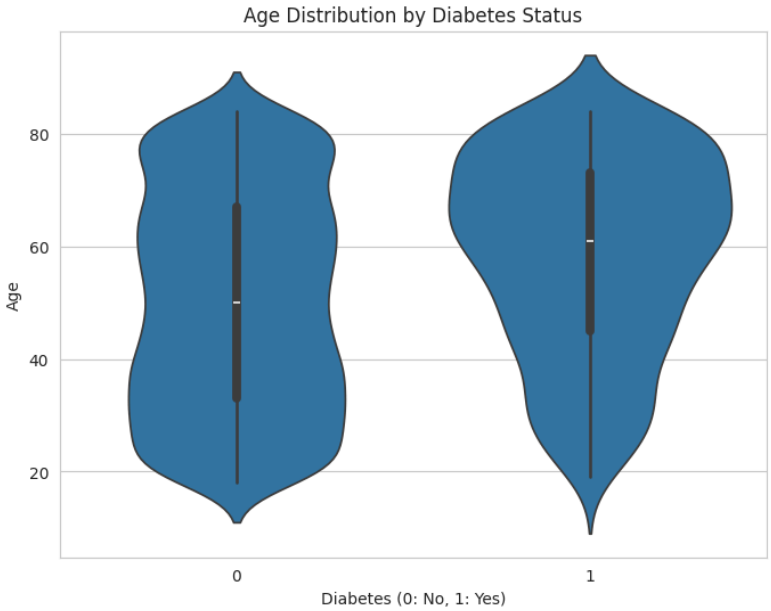
- **Clean Data:** With **0 missing values**, models utilize a complete feature set.
- **Imbalance:** Only **9.03%** (542) have diabetes, requiring models to be sensitive to the minority class.
- **Demographics:** Ages range from **18–84** (mean **51**), representing diverse life stages.
- **Health Metrics:** Average **BMI is 26.0** (overweight) and blood pressure is **134.2 mmHg** (near Stage 1 hypertension).

Analysis of the Distributions

1. The Role of Age in Classification

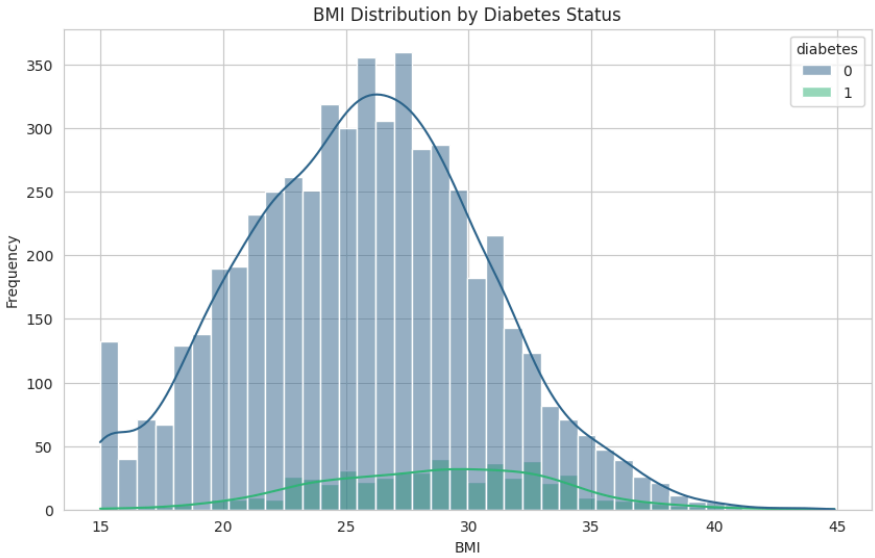
The "Age Distribution by Diabetes Status" plot shows that while non-diabetic patients are spread relatively evenly across all age groups, the diabetic population is concentrated in the older

demographics (roughly **60 – 80 years**). This confirms age as a vital feature for the classification task.



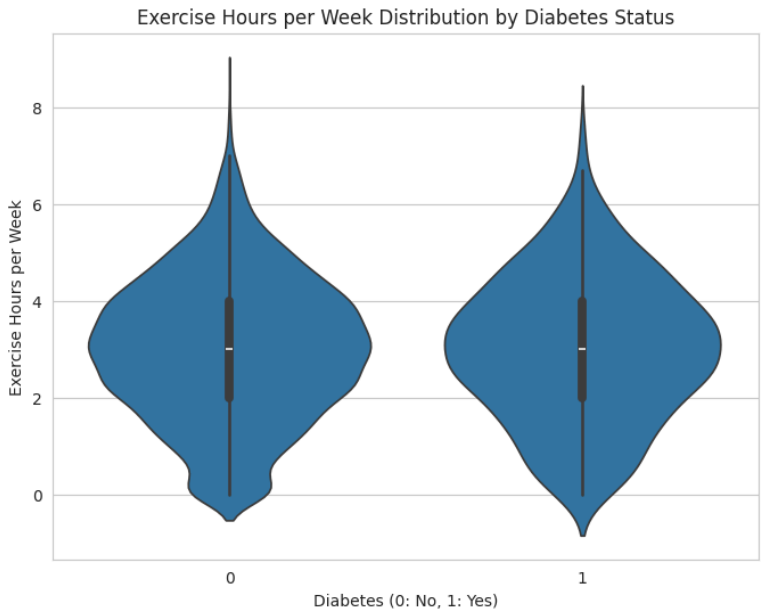
2. BMI Distribution Analysis

The "BMI Distribution" chart illustrates a distinct shift between the two groups. While the "Diabetes: 0" curve (blue) peaks earlier, the "Diabetes: 1" curve (green) shows a later peak. This demonstrates that although many individuals with a BMI of 25 remain non-diabetic, the **likelihood** of a diabetes diagnosis rises significantly alongside increasing BMI, specifically once it surpasses the **30.0** threshold.



3. Analysis of Physical Activity Levels

The "Exercise Hours per Week Distribution by Diabetes Status" visualization reveals that the distribution for individuals with **Diabetes: 1** is notably denser at the lower end of the scale compared to those with **Diabetes: 0**. This indicates that, on average, patients diagnosed with diabetes tend to report **fewer exercise hours** per week.



Important Features Influencing Prediction

Distribution plots highlight variables with the most significant variance between diabetic and non-diabetic individuals.

- **BMI:** The strongest predictor (correlation **0.17**). Diabetic density is significantly higher in the **30.0 to 35.0** range compared to the non-diabetic peak at 25.
- **Blood Pressure:** The second most impactful feature (correlation **0.15**), reflecting its strong link to metabolic issues.
- **Age:** Diabetic cases are concentrated among older individuals, specifically between **60 and 80 years old**, with a **0.12** correlation.

3. Model Implementation

Explanation of the Machine Learning Model

The traditional Machine Learning model used is **Logistic Regression**. This is a linear classification algorithm used for binary prediction that calculates the probability of the target class based on a weighted sum of input features (patient health attributes). The coefficients derived from training indicate both the strength and direction of a feature's influence on the diabetes prediction.

Explanation of the Neural Network Architecture

The Deep Learning approach uses a **Sequential Artificial Neural Network (ANN)** designed for binary classification.

- **Input Layer:** Accepts 5 normalized features: Age, BMI, Blood Pressure, Smoker Status, and Exercise Hours.
- **Hidden Layers:** Consists of two fully connected (Dense) layers with **16 and 8 neurons** respectively. The **ReLU activation** function is used in both to introduce non-linearity.
- **Output Layer:** A single neuron with a **Sigmoid activation** function, which outputs the final probability score between 0 and 1.

4. Model Evaluation

Performance Metrics for Both Models

Traditional ML Model (Logistic Regression)

The model achieved an overall **Accuracy of 65%**. Given the class imbalance, this accuracy reflects an attempt to prioritize the identification of the minority (diabetic) class.

Feature Importance:

The model's coefficients quantify the influence of each feature on the prediction:

- **BMI (0.524):** Most significant positive predictor.
- **Age (0.337):** Second most influential positive predictor.
- **Blood Pressure (0.273):** Primary physiological marker.
- **Smoker (-0.176):** Shows a small inverse relationship.
- **Exercise Hours (0.033):** Least influential feature.

Deep Learning Model (ANN)

The model achieved a high overall **Accuracy of 90.5% (0.9050)** with a **Loss of 0.3473** on the test data. However, the performance on the minority class (Diabetes=1) is critically low.

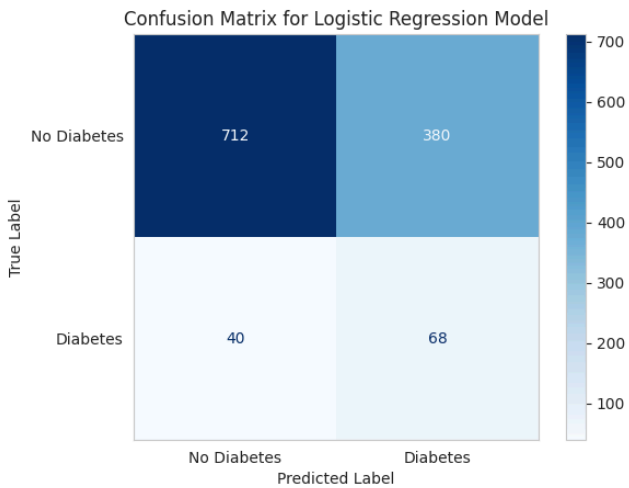
Visualizations (Confusion Matrix and Classification Report)

Logistic Regression Classification Report

```
Classification Report for ANN:
              precision    recall  f1-score   support
```

0	0.91	0.99	0.95	1092
1	0.29	0.04	0.07	108
accuracy			0.91	1200
macro avg	0.60	0.51	0.51	1200
weighted avg	0.86	0.91	0.87	1200

Logistic Regression Confusion Matrix



ANN Classification Report

Classification Report for ANN:

	precision	recall	f1-score	support
0	0.91	0.99	0.95	1092
1	0.29	0.04	0.07	108
accuracy			0.91	1200
macro avg	0.60	0.51	0.51	1200
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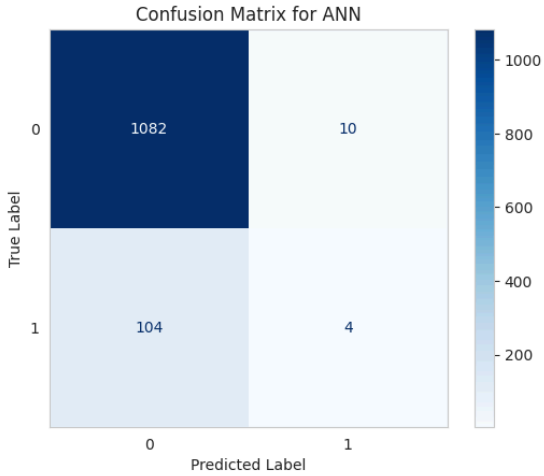
ANN Confusion Matrix:



Post SMOTE Analysis

After implementing the SMOTE Technique to handle imbalanced datasets by generating synthetic samples for the minority class, rather than just duplicating them. The training set has an equal number of "Non-Diabetic" and "Diabetic" cases.
This Increases the Logistic Regression model and ANN model's accuracy and recall.
Here are the classification report after implementing SMOTE Method:

```
--- Logistic Regression with SMOTE Results ---

              precision    recall  f1-score   support

           0       0.95      0.64      0.77      1092
           1       0.15      0.63      0.24       108

    accuracy                           0.64      1200
   macro avg       0.55      0.64      0.50      1200
weighted avg       0.87      0.64      0.72      1200

--- ANN with SMOTE Results ---

               precision    recall  f1-score   support

           0       0.93      0.77      0.84      1092
           1       0.14      0.38      0.20       108

    accuracy                           0.73      1200
```

```
   macro avg       0.53      0.57      0.52      1200
weighted avg       0.86      0.73      0.78      1200
```

5. Comparative Analysis

Differences Between the Models

The primary distinction lies in the architectural approach and resulting performance profile:

- **Overall Performance vs. Specificity:** The Deep Learning (ANN) model achieved a significantly higher overall accuracy (**90.5%**) than the traditional Logistic Regression model (**65%**). However, this high accuracy for the ANN is misleading, as the Logistic Regression model, being linear, provides superior interpretability through explicit feature coefficients.
- **Feature Influence:** Logistic Regression clearly ranks the features, identifying **BMI (0.524)**, **Age (0.337)**, and **Blood Pressure (0.273)** as the strongest positive predictors of risk. The non-linear nature of the ANN (using ReLU activation) makes feature importance opaque.
- **Model Complexity:** The Logistic Regression is a simple, linear model, while the ANN is a non-linear, multi-layer architecture.

Reasons for Accuracy Differences

The substantial difference in overall accuracy is a direct consequence of the **severe class imbalance** (90.97% non-diabetic vs. 9.03% diabetic) and the models' response to this imbalance:

- **ANN Overfitting to Majority Class:** The ANN's impressive overall accuracy (90.5%) is achieved by correctly classifying the vast majority of non-diabetic patients. This is evidenced by the exceptionally high Recall for the non-diabetic class (**0.99**). Conversely, the model effectively ignores the minority class, resulting in a critically low Recall for diabetic patients (**0.04**). The model is therefore highly biased towards predicting "No Diabetes."
- **Logistic Regression's Balanced Approach:** The lower accuracy of the Logistic Regression model (**65%**) suggests it established a more conservative decision boundary. While its overall performance is inferior, its less aggressive classification strategy may offer a more balanced, though still imperfect, trade-off between False Positives and False Negatives than the highly skewed ANN.

Recommendations for Real-World Use

For a healthcare risk prediction application, the cost of a False Negative (failing to identify an at-risk patient) is far greater than the cost of a False Positive. Therefore, the current models require refinement:

- **Current ANN is Clinically Unsuitable:** An ANN model with a **Recall of 0.04** on the diabetic class means it misses 96% of patients who actually have the condition, rendering it useless for clinical screening.
- **Prioritize Recall over Accuracy:** Any subsequent model training must focus on optimizing metrics that prioritize the minority class, such as the F1-score or Recall for the positive class (Diabetes=1).
- **Recommendation for Deployment:** The **Logistic Regression model** is immediately preferable for its **interpretability** in a clinical context. However, both models require **re-training with robust techniques** to address the class imbalance (e.g., using class weights, synthetic oversampling like SMOTE, or adjusting the probability threshold) to achieve an acceptable level of sensitivity (Recall) for the diabetic population.